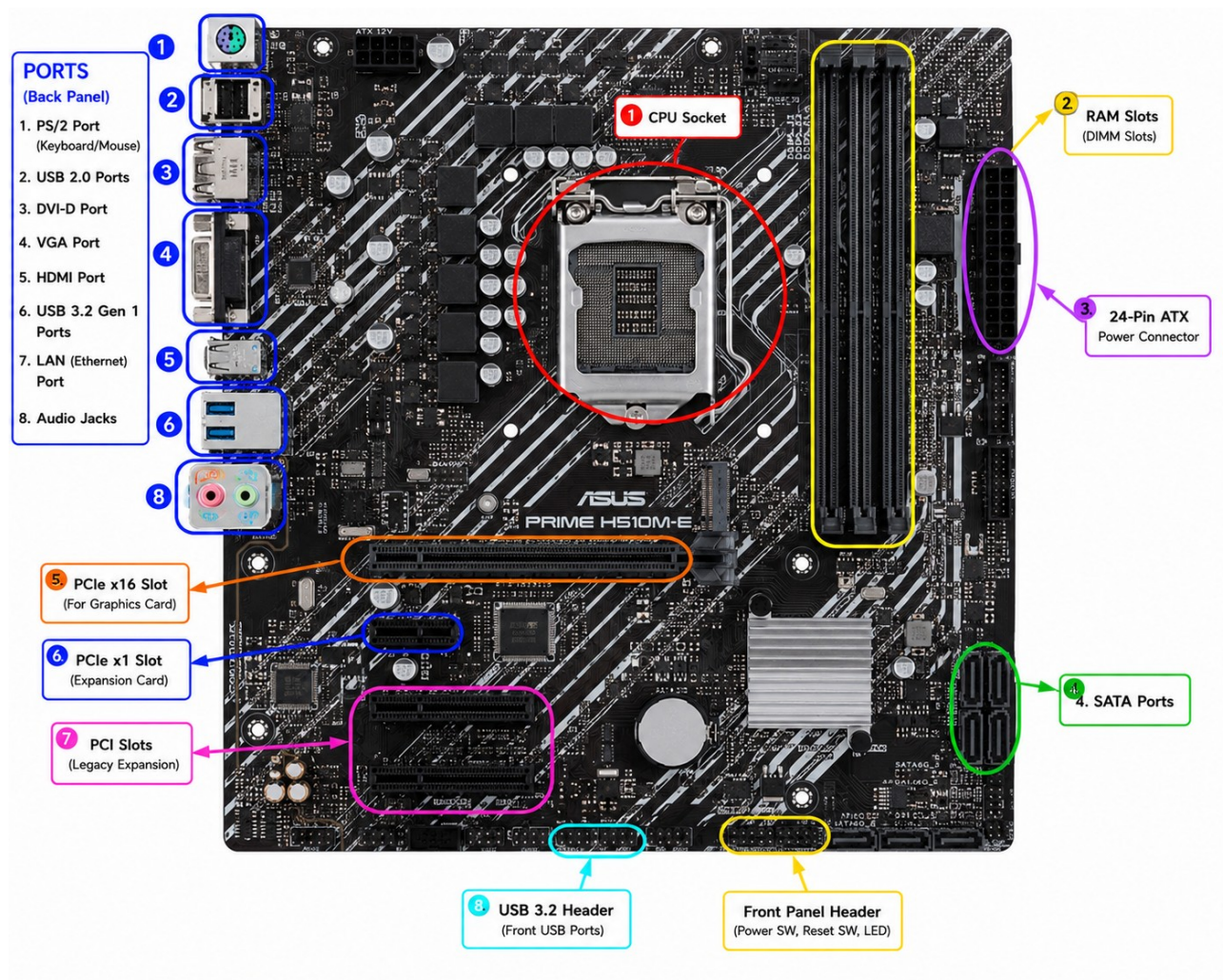


1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

CPU Socket Type	Manufacturer	Compatible CPU Model
LGA1700	Intel	Intel Core i5-12600K
AM5	AMD	AMD Ryzen 7 7700X
sTRX4	AMD	AMD Ryzen Threadripper 3970X

3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

3. CPU and Chipset Functions

CPU (Central Processing Unit)

The CPU is the main processor of the computer. It executes instructions, performs calculations, and controls software operations. It is often called the "brain" of the computer.

Chipset

The chipset manages communication between the CPU and other hardware components such as RAM, storage drives, USB devices, network adapters, and graphics cards. It acts like a traffic controller on the motherboard.

How They Work Together When Opening Spotify

When you open Spotify:

1. The CPU receives the command to launch the application.
2. The chipset helps retrieve Spotify files from the SSD or HDD.
3. The chipset transfers data between storage, RAM, and the CPU.
4. The CPU processes the program instructions.
5. The GPU displays the Spotify interface on the monitor.
6. The chipset manages audio, USB devices, and network communication while Spotify streams music.

Together, the CPU performs the processing while the chipset coordinates data movement between hardware components.

4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

